

ALBERT OLAFARE

Embedded Systems Engineer | Electronics Designer | IoT Hardware Specialist

PROFESSIONAL SUMMARY

Embedded Systems Engineer with 4+ years of end-to-end hardware development experience, specializing in IoT product design — from schematic capture and custom PCB layout through embedded C/C++ firmware development and field deployment. Delivered 20+ production-ready systems including WiFi-enabled controllers, safety-critical detection devices, and battery-powered IoT sensors. Known for translating early-stage concepts into reliable, manufacturable hardware that performs in real-world conditions, with hands-on expertise spanning hardware design, firmware engineering, and mechanical enclosure development.

PROFESSIONAL EXPERIENCE

Embedded Systems Engineer & Hardware Designer 01/2022 – Present Independent / Freelance – Abuja, Nigeria (Remote)

- Designed 20+ custom PCBs for IoT and automation applications using KiCad, managing the complete development cycle from requirements gathering through prototype validation and manufacturing handoff.
- Produced JLCPCB-ready manufacturing packages (Gerbers, BOMs with LCSC part numbers, pick-and-place files) enabling seamless board fabrication.
- Implemented power management systems for battery-operated IoT devices, including sleep-cycle optimization and charge management circuitry.
- Designed high-voltage switching circuits (220V AC) with proper galvanic isolation, safety features, and EMI mitigation.
- Developed embedded C/C++ firmware for ESP32, STM32, and ATmega microcontrollers with emphasis on stability, performance, and maintainability.
- Integrated sensors via I²C, SPI, and UART with robust error handling, data validation, and calibration routines.
- Built WiFi-connected features including MQTT communication, OTA firmware updates, and cloud platform integration.
- Implemented real-time control systems using FreeRTOS, interrupt-driven architectures, and state machine logic.
- Created 3D enclosures in SolidWorks with thermal management, EMI shielding, and DFM considerations baked in from the start.
- Produced photorealistic product renders in Blender for client presentations and pre-production marketing.

Hardware Engineering Intern TD4PAI Hardtech Hub – FCT, Abuja, Nigeria

Jun 2025 – Dec 2025

- Contributed to PCB design and prototype assembly for IoT sensor devices within a multidisciplinary product development team.
- Identified and implemented Creality Print software for wireless 3D printer control, eliminating the USB drive workflow and enabling real-time monitoring — adopted hub-wide.
- Optimized 3D printing workflows including material selection (ABS, PLA, Nylon), temperature tuning, and post-processing.
- Designed and iteratively improved mechanical enclosures for hub projects through functional prototyping and testing.

CONTACT

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WEBSITES & PROFILES

- linkedin.com/in/albertOla
- alberthycntte.vercel.app

TECHNICAL SKILLS

Embedded Development

- C, C++, Python
- ESP32 / ESP8266, STM32 (ARM Cortex-M), ATmega
- FreeRTOS, interrupt-driven design, state machines
- I²C, SPI, UART, MQTT, HTTP/HTTPS, WiFi, BLE

Hardware Design

- KiCad — schematic capture, layout, DRC, Gerbers
- Analog/digital circuits, power management
- AC load control (220V), relay switching
- Battery management, sensor interfaces
- JLCPCB-ready files (BOM, CPL), DFM

Mechanical & CAD

- SolidWorks (3D modeling, assemblies, drawings)
- Blender (photorealistic rendering, animation)
- 3D printing — FDM (ABS, PLA, Nylon)
- Tinkercad, rapid prototyping

Dev Tools & Testing

- VS Code, Arduino IDE, ESP-IDF, STM32CubeIDE
- Git, GitHub
- Oscilloscope, logic analyzer, multimeter

- Gained hands-on exposure to design-for-manufacturing (DFM) principles and production-readiness review processes.

EDUCATION

B.Tech in Physics Electronics

Expected Aug 2026

Federal University of Technology Akure (FUTA) · Ondo State, Nigeria

Relevant Coursework: Digital Electronics · Microprocessor Systems · Embedded Systems Design · Power Electronics · Control Systems · Analog Circuit Design · Sensor Physics · Communication Systems

CERTIFICATIONS

- Intro to Electronics Simulation — Alison (2021)
- Intro to ESP32 Web Server — Alison (2021)
- Arduino Programming — Temosan Tech (2020)
- Elementary Electronics — Temosan Tech (2020)

AREAS OF INTEREST

- IoT Device Development
- Audio & Sensor Systems
- Battery-Powered Electronics
- Wireless Communication
- Embedded AI / TinyML
- Edge Computing
- Safety-Critical Systems
- Smart Agriculture & Industrial Automation